

# Efficient gate-set extensions for fault-tolerant qudit quantum computation

George T. Fleming,<sup>1</sup> Henry Lamm<sup>1,\*</sup>, Karla Tame-Narvaez,<sup>1</sup> and Peter Vander Griend<sup>1,2</sup>

<sup>1</sup>*Fermi National Accelerator Laboratory, Batavia, IL 60510, USA*

<sup>2</sup>*Department of Physics and Astronomy, University of Kentucky, Lexington, KY 40506, USA*

(Dated: April 14, 2026)

Fault-tolerant quantum computation requires a finite universal gate set whose members have cheap physical realizations. We introduce SWIFTBOT, an agentic workflow that numerically ranks finite subgroups of  $SU(d)$  extended by a single qudit gate, using the Quantum Circuit Overhead  $Q_T$  as the figure of merit. At  $d = 3$  we produce the first systematic  $Q_T$  ranking across three  $\Sigma$ -series subgroups, finding that Haar-random extensions on the largest approach the Kesten–McKay floor within 32%, while structured diagonal and Howard–Vala extensions sit factors of 3–70 above it. Applied to gate sets relevant for lattice-gauge theory simulation, a non-canonical rational-phase extension delivers  $6.8\times$  lower mean synthesis depth than canonical Clifford+ $T$ , demonstrating that the best general-purpose extension and the best target-specific extension can differ by nearly an order of magnitude. The pipeline, registered groups, fault-tolerance catalog, and reproduction data are released with the paper.

## I. INTRODUCTION

Fault-tolerant quantum computation requires a finite universal gate set whose members have cheap physical realizations. The tension between these requirements is sharp: the Eastin–Knill theorem [1] forbids any nontrivial stabilizer code from admitting a universal *transversal* gate set. Practical designs therefore split the gate set into a transversal subgroup  $\mathcal{C} \subset SU(d)$  (usually a finite subgroup the error-correcting code protects cheaply) and one or more non-transversal extension gates  $g$  made fault-tolerant by magic-state distillation [2, 3] or by code switching between codes with complementary transversal structure. Each extension gate  $g$  comes with its own cost: a dedicated magic state, a code that supports it transversally, a distillation protocol, or some combination of the three.

On qubits this tension has a standard resolution. Clifford+ $T$  [2, 4] pairs the Clifford group with  $T = \text{diag}(1, e^{i\pi/4})$ ; Clifford gates are transversal on the Steane, Reed–Muller, and Bravyi–Haah families, and  $T$  is distilled via Bravyi–Kitaev [2] or Bravyi–Haah [3]. Ross–Selinger [5] and Kliuchnikov–Maslov–Mosca [6] make single-qubit synthesis into Clifford+ $T$  a solved problem with certified  $T$ -count bounds. For  $d = 2$  the choice of gate set is effectively fixed.

No comparable consensus exists at  $d > 2$ . Howard and Vala [7] proposed a three-parameter family  $HV(z, \gamma, \varepsilon)$  of qudit  $T$ -gates at prime  $d$ , but no systematic comparison of members has been done. The qudit finite-subgroup landscape is known [8, 9], transversal-gate classifications for stabilizer codes have been cataloged [10], and Campbell–Anwar–Browne give prime- $d$  distillation protocols [11]. What is missing is a quantitative ranking: given a base group  $\mathcal{C}$  and an extension  $g$ , how efficiently does  $(\mathcal{C}, g)$  synthesize target unitaries?

Ślowski, Dulián, and Sawicki [12] supplied the right figure of merit. The *Quantum Circuit Overhead*  $Q_T(\mathcal{C}, g)$  is the asymptotic exponent in the word length of an  $\varepsilon$ -accurate approximation: the number of gates scales as  $(1/\varepsilon)^{Q_T}$ , bounded below by the Kesten–McKay optimum  $Q_{\text{opt}}$  and saturated by the “golden gates” of Sarnak [13].

This paper uses  $Q_T$  to rank candidates. We introduce SWIFTBOT, an agentic workflow that enumerates  $(\mathcal{C}, g)$  pairs at  $d = 2, 3, 4$ , screens each for universality via Sawicki’s criterion, evaluates  $Q_T$  numerically [12], and cross-references surviving pairs against a curated fault-tolerance catalog. With it we make three contributions. First, at  $d = 2$  we reproduce the Ślowski *et al.* numerics at the percent level (Section IV). Second, we establish the first paper-scale  $Q_T$  ranking at  $d = 3$  across the three irreducible-adjoint  $\Sigma$ -series subgroups with  $|\mathcal{C}| \in \{216, 648, 1080\}$ , using Howard–Vala, rational-phase, and Haar-random extensions (Section V). Third, applied to a gate set relevant for discrete-subgroup lattice gauge simulation [14–16], a non-canonical rational-phase extension beats Clifford+ $T$  by  $6.8\times$  in mean synthesis depth (Section VI)—showing that the best general-purpose extension and the best target-specific extension can differ by nearly an order of magnitude, a distinction invisible when  $Q_T$  alone is reported.

Code and data are released with this paper [17]. Section II reviews the QCO framework, Section III sketches the pipeline, Sections IV to VI present the results, and Section VII closes with a fault-tolerance audit and open problems.

## II. BACKGROUND

We briefly review the three ingredients used throughout: the Quantum Circuit Overhead  $Q_T$  [12] as our primary figure of merit, the Sawicki criterion [18] for ruling out non-universal extensions without running QCO, and the Howard–Vala family [7] and DSA primitive-gate series [14, 15] as the principal sources of candidate extensions at

\* hlamm@fnal.gov

$d \geq 3$ .

### II.1. $\delta$ -approximate $t$ -designs and $Q_T$

Let  $\mathcal{S} = \mathcal{C} \cup \mathcal{C}^{-1} \cup \{g, g^{-1}\} \subset \text{SU}(d)$  be a symmetric generating set consisting of a finite subgroup  $\mathcal{C}$  and a single extension  $g$ . Consider the  $t$ -design operator on the  $t$ -th tensor power of the defining representation,

$$T_t(\mathcal{S}) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \pi_t(s), \quad \pi_t = \bigoplus_{\lambda \in \Lambda_t} \pi_\lambda^{\oplus m_\lambda}, \quad (1)$$

where  $\Lambda_t$  is the set of  $\text{SU}(d)$  highest weights that appear in the decomposition of  $(\mathbb{C}^d)^{\otimes t} \otimes (\mathbb{C}^d)^{* \otimes t}$  and  $\pi_\lambda$  is the associated irreducible representation. The  $t$ -design spectral radius is

$$\delta_t(\mathcal{S}) = \max_{\lambda \in \Lambda_t \setminus \{0\}} \|T_t(\mathcal{S})\|_{\pi_\lambda}, \quad (2)$$

the largest spectral norm across non-trivial irreps. A smaller  $\delta_t$  means the random walk generated by  $\mathcal{S}$  equidistributes faster on  $\text{SU}(d)$ .

For an  $\varepsilon$ -net of  $\text{SU}(d)$  in the operator norm, Słowiak *et al.* [12] show that the number of Solovay–Kitaev-style words required scales as  $(1/\varepsilon)^{Q_T}$ , with

$$Q_T(\mathcal{S}) = \frac{\log |\mathcal{C}|}{\log(\delta_t^{-1}(\mathcal{S}))}, \quad (3)$$

in the limit  $t \rightarrow \infty$ . The Kesten–McKay bound for a  $k$ -regular Cayley graph gives

$$\delta_t \geq \delta_{\text{opt}} = \frac{2\sqrt{|\mathcal{C}| - 1}}{|\mathcal{C}|}, \quad (4)$$

and therefore

$$Q_T \geq Q_{\text{opt}} = \frac{\log |\mathcal{C}|}{\log(1/\delta_{\text{opt}})}, \quad (5)$$

with equality for “golden gates” [13]. Representative values:  $Q_{\text{opt}} \approx 3.47$  for the qubit Clifford group, 4.19 for the projective  $A_4 \subset \text{PU}(2)$  (registered as `hurwitz`), 2.81 for the binary icosahedral group  $2I$  (BI), and 2.69 for the smallest  $d = 3$   $\Sigma$ -series group  $\Sigma(72 \times 3)$ .

In practice  $Q_T$  is estimated at finite  $t$  by sampling the ensemble in Eq. (1) with a Monte-Carlo extension gate  $g$  (the *rnd* baseline) or a deterministic one, then evaluating Eq. (2) via sparse spectral decomposition on each irrep of  $\pi_t$ . The feasibility ceiling at  $d = 3$  is set by representation dimensions [12]: the largest irrep at  $t = 50$  has dimension  $\sim 10^5$ , whereas  $t = 15$  peaks at  $\sim 4 \times 10^3$  and is tractable. Throughout this work we take  $t$  large enough that the estimate of  $Q_T$  has stabilised within a few percent, and we quote  $Q_{\text{opt}}$  alongside  $Q_T$  to expose the gap to optimality.

### II.2. Pre-screening with the Sawicki criterion

Not every candidate extension needs a QCO run. Sawicki and Karnas [18] proved that  $\mathcal{S}$  generates a dense subgroup of  $\text{SU}(d)$  if and only if

- (i) the adjoint representation of  $\mathcal{S}$  is irreducible, and
- (ii)  $\mathcal{S}$  is not contained in a maximal finite subgroup of the centralizer.

Let  $\text{Comm}(\text{Ad}_{\mathcal{C}})$  denote the space of complex matrices that commute with every  $\text{Ad}_c$  for  $c \in \mathcal{C}$ ; its dimension is 1 if and only if the adjoint representation of  $\mathcal{C}$  is irreducible. Condition (i) is then checkable in  $\mathcal{O}(d^2)$  memory via Schur’s orthogonality relation,

$$\dim \text{Comm}(\text{Ad}_{\mathcal{C}}) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} |\chi_{\text{Ad}}(c)|^2, \quad (6)$$

where  $\chi_{\text{Ad}}(c) = \text{tr Ad}_c$ . A value  $> 1$  means the adjoint splits, and no extension preserving that splitting can be universal. This immediately excludes from consideration, e.g.,  $\Sigma(36 \times 3)$  (the 108-element crystallographic group), the 60-element simple group  $\Sigma(60) \simeq A_5$ , and the  $s_{60 \times 4}$  family at  $d = 3$ , since any diagonal extension preserves the reducible-adjoint structure. Condition (ii) is enforced at finite cost by running a bounded closure  $\langle \mathcal{S} \rangle \cap \text{Cap}(n)$  against the classification of finite subgroups of  $\text{SU}(d)$  (orders  $\leq 120$  at  $d = 2$ ,  $\leq 1080$  at  $d = 3$ ,  $\leq 5040$  at  $d = 4$ ) [8, 9]; we use cap sizes  $n = 120, 1080, 5040$  respectively. An extension is flagged `finite` if closure terminates below the cap, `universal_likely` if closure survives the cap and condition (i) holds, and `infinite_but_not_dense` if condition (i) fails but closure is open.

### II.3. Candidate extensions

At  $d = 2$  the canonical extension is  $T = P(\pi/4) = \text{diag}(1, e^{i\pi/4})$ , giving the Clifford+ $T$  set with certified synthesis via Ross–Selinger [5] and Kliuchnikov–Maslov–Mosca [6]. We also consider diagonal  $P(\theta)$  for  $\theta \in \{\pi/8, 2\pi/9, \pi/9, \pi/18\}$ , the Parzanchevski–Sarnak supergolden gates [19], and Haar-random extensions.

At prime  $d$  the Howard–Vala family [7] generalizes  $T$  via three integer parameters  $(z, \gamma, \varepsilon)$ :

$$\text{HV}(z, \gamma, \varepsilon) = \mathcal{N} \cdot \text{diag}(\zeta_d^{e_0}, \zeta_d^{e_1}, \dots, \zeta_d^{e_{d-1}}), \quad (7)$$

with  $\zeta_d = e^{2\pi i/d}$ , exponents  $e_k = zk^3 + 3\gamma k^2 + \varepsilon$ , and  $\mathcal{N}$  a phase chosen so that  $\det \text{HV}(z, \gamma, \varepsilon) = 1$ . Our panel uses  $(z, \gamma, \varepsilon) \in \{(1, 1, 0), (1, 2, 0), (2, 1, 0)\}$ : the first is Campbell’s magic gate [11], the second is the Howard–Vala Eq. (27) variant (Clifford-equivalent at  $d = 3$ ), and the third is a representative new orbit.

Discrete-subgroup approximations of lattice gauge simulations for  $\text{SU}(d)$  [14, 15] also pick out a different kind of target. For each chosen qudit group, the continuous

degree of freedom reduces to a trace rotation  $U_{\text{Tr}}(\theta)$  that decomposes into a finite set of  $R_Z(\alpha_c \theta)$  rotations whose coefficients  $\alpha_c$  depend on the specific subgroup and encoding. This motivates a *target-family* figure of merit: the mean breadth-first-search depth to synthesize the required  $R_Z$  rotations within  $\varepsilon \leq 10^{-2}$ . We instantiate it for one specific subgroup in Section VI.

### III. SWIFTBOT PIPELINE

SWIFTBOT is a five-stage agentic pipeline built around the QCO kernel of [12]. Numerical stages are deterministic; an LLM supervisor intervenes only at stage boundaries to rank candidates, propose extensions, and decide when to stop. All headline numbers below come from deterministic panels that bypass the LLM; the agent is used for exploratory search not reported here. Full code and a command-line driver are released with the paper [17].

*Stage 1 (group registration)* computes projective closures  $\langle \mathcal{G} \rangle \subset \text{SU}(d)/\text{Z}(\text{SU}(d))$  from seed unitaries and stores each group under a content-addressable key. The registry covers the canonical  $d \in \{2, 3, 4\}$  subgroups used here: Pauli–Clifford, the projective  $A_4 \subset \text{PU}(2)$  (registered as *hurwitz*), and  $2I$  (BI) at  $d = 2$ ; the  $\Sigma(m \times 3)$  series for  $m \in \{36, 72, 216, 360\}$  at  $d = 3$ ; and selected Fairbairn subgroups at  $d = 4$ .

*Stage 2 (universality screen)* evaluates  $\dim \text{Comm}(\text{Ad}_C)$  via Eq. (6) and runs a bounded closure against the classification caps  $\{120, 1080, 5040\}$  [8, 9]. Pairs whose closure terminates below cap are tagged *finite* and skipped; reducible-adjoint pairs are tagged diagnostically.

*Stage 3 (QCO numerics)* invokes the kernel of [12] as a subprocess. Two optimizations help: (i) every gate has the form  $gcg^\dagger$  with  $c \in \mathcal{C}$ , so  $\pi_\lambda(gcg^\dagger) = \pi_\lambda(g)\pi_\lambda(c)\pi_\lambda(g)^\dagger$  replaces  $|\mathcal{C}|$  exponentials with one; (ii) the  $\pi_\lambda(c)$  lists are persisted to disk keyed by a fingerprint of  $\mathcal{C}$  and the highest weight  $\lambda$ . Together these give 3–6 $\times$  speedup at  $d = 3$ . Results are stored as per-sample QTRecords in a SQLite cache with full provenance.

*Stages 4–5 (fault-tolerance)* match each surviving  $(\mathcal{C}, g)$  against a curated catalog of QEC codes (Steane, Reed–Muller, Bravyi–Haah, Kubischta 2I, qutrit triorthogonal, ternary Golay, Pollatsek–Ruskai PI, Kubischta–Teixeira PI, and more) and distillation protocols (Bravyi–Kitaev, Bravyi–Haah, Campbell–Anwar–Brown, programmable MSD, OJB code-switching). Unmatched pairs return a *research-needed* marker with an EC Zoo [20] pointer rather than guessing.

### IV. VALIDATION AT $d = 2$

We validate the SWIFTBOT QCO kernel against the  $d = 2$  reference dataset of Slowik, Dulián, and Sawicki [12] on two qubit bases: the Clifford group ( $|\mathcal{C}| = 24$ ) and the 12-element projective  $A_4 \subset \text{PU}(2)$  (registered *hurwitz*),

each paired with  $P(\pi/4)$  and with Haar-random extensions. We also report a pilot on the binary icosahedral group  $2I$  (BI,  $|\mathcal{C}| = 120$ ), which carries the Kubischta nonadditive transversal code [21]. Runs are at  $t = 50$ , matching the reference. A zeroth-order check—replaying the  $\delta$ -parser on all 5050 rows of the reference file and recomputing  $Q_T$ —matches the reference to  $\mathcal{O}(10^{-15})$ , isolating any later disagreement to the spectral kernel. Headline results are summarized in Table I: Clifford+ $P(\pi/4)$  yields  $Q_T = 51.9$  (reference  $\approx 52$ ); Haar-random extensions agree at the few-percent level. A paper-scale rerun at  $10^4$  samples on BI is currently in execution and will supersede the pilot in the final version.

*Two remarks.* First, Clifford+ $T$  is about  $14\times$  worse than Haar-random on  $Q_T$  (51.9 vs 3.79)—commensurability between  $P(\pi/4)$  and the Clifford orbit slows the Haar random walk by more than an order of magnitude in the approximation exponent [12]. This is a  $Q_T$  claim, not a verdict on Clifford+ $T$  itself: when the target is structured (Section VI), the ranking inverts. Second, the pilot BI ratio  $Q_T/Q_{\text{opt}} \approx 1.36$  is far closer to saturation than Clifford’s and arrives with a cataloged fault-tolerant code [21], motivating the sample\_size =  $10^4$  rerun.

### V. $d = 3$ SURVEY

With the SWIFTBOT kernel validated at  $d = 2$  (Section IV), we turn to the first paper-scale  $Q_T$  ranking at  $d = 3$ . The survey covers the three irreducible-adjoint  $\Sigma$ -series subgroups of  $\text{SU}(3)$ — $\Sigma(72 \times 3)$ ,  $\Sigma(216 \times 3)$ , and  $\Sigma(360 \times 3)$ —each extended by a common panel of eight candidate gates. The smaller groups  $\Sigma(36 \times 3)$  (the 108-element crystallographic Valentiner analogue) and  $\Sigma(60) \simeq A_5$  have reducible adjoint representations (cf. Eq. (6)) and are excluded at Stage 2 of the pipeline: no extension preserving the invariant subspace structure of  $\text{su}(3)$  can render them universal, so their  $Q_T$  is formally divergent [18].

#### V.1. Panel specification

The panel fixes eight extensions per base group, with identical parameters across the three groups. Three are Howard–Vala variants  $\text{HV}(1, 1, 0)$ ,  $\text{HV}(1, 2, 0)$ ,  $\text{HV}(2, 1, 0)$  (see Eq. (7)). Three are diagonal phase gates  $P(\theta)$  at angles  $\theta \in \{2\pi/9, \pi/9, 2\pi/5\}$ , chosen to span both the cyclotomic-9th-root sector relevant to Section VI and the coarser  $2\pi/5$  angle that generates no low-order closure against any of the three base groups. The remaining two slots per group are independent Haar-random batches of size 10, drawn freshly per subprocess from the invariant measure on  $\text{SU}(3)$ ; these serve as the *rnd* baseline of [12]. All eight extensions survive the Stage 2 universality screen of Section II.2: each  $\Sigma$ -series base has one-dimensional

TABLE I.  $d = 2$  validation of the SWIFTBOT QCO kernel against the reference dataset of Słowik, Dulián, and Sawicki [12]. All values computed at  $t = 50$ ; the Haar-random (rnd) rows report the best  $Q_T$  across the stated number of samples drawn from the invariant measure on  $SU(2)$ ; the fixed-diagonal row is deterministic. The  $Q_{\text{opt}}$  column is the Kesten–McKay floor of Eq. (5). Reference values are the  $t \rightarrow \infty$  headline numbers of [12] when available; BI has no published QCO reference.

| Base ( $ \mathcal{C} $ ) | Extension                  | $Q_T$ (this work) | $Q_T$ [12]                   | $Q_{\text{opt}}$ |
|--------------------------|----------------------------|-------------------|------------------------------|------------------|
| Clifford (24)            | $P(\pi/4)$ , canonical $T$ | 51.91             | $\approx 52$                 | 3.47             |
| Clifford (24)            | Haar-rnd (best of 20)      | 3.79              | $\approx 3.7$                | 3.47             |
| hurwitz (12)             | Haar-rnd (best of 20)      | 4.34              | $\approx 4.0$ (super-golden) | 4.19             |
| BI (120)                 | Haar-rnd (best of 200)     | 3.83              | —                            | 2.81             |

adjoint commutant by Eq. (6), and the bounded closures stay open past the  $|\mathcal{C}| \leq 1080$  cap of [8, 9].

The QCO numerics run at  $t = 5$ , the largest design order for which the full panel completes on a single 32-core workstation within the budget of an interactive session. Fixed extensions use `sample_size = 1` (the gate is deterministic); Haar-random batches use `sample_size = 10`. Each (group, extension) pair is dispatched as an independent `qco` subprocess with `OMP_NUM_THREADS=1` and `--workers 8` to keep the per-irrep spectral evaluation of Eq. (2) single-threaded while exploiting thread-level parallelism across the  $|\mathcal{C}|$  matrix actions described in Section III. The full Tier-1 panel—3 groups  $\times$  8 extensions, with the cached  $\pi_\lambda(\mathcal{C})$  tables rebuilt once per group—completes in approximately 17 minutes of wall time. All numerical values reported below are reproduced bit-identically from the SQLite cache released with the paper [17].

## V.2. Tier-1 ranking

Table II lists every (group, extension) pair, ranked within each base group by best  $\delta$  ascending. The companion column  $Q_T / \log |\mathcal{C}|$  removes the  $|\mathcal{C}|$  scale from Eq. (3) and exposes the spectral-gap content of the extension itself.

Three features stand out in Table II and Fig. 1. First, five of six structured extensions give *identical*  $\delta$  to four decimals across all three groups; only  $HV(2, 1, 0)$  separates  $\Sigma(360 \times 3)$ . The shared dominant spectral direction at  $t = 5$  lives in a diagonal subalgebra common to the  $\Sigma$ -series, and  $|\mathcal{C}|$  alone does not sharpen it—subdominant irreps at  $t \geq 10$  may break the degeneracy (Section V.5). Second, because  $\delta$  is essentially constant while  $\log |\mathcal{C}|$  grows, the structured extensions’  $Q_T$  grows with  $|\mathcal{C}|$ : not because smaller groups are better, but because a larger group needs proportionally more depth to densely cover  $SU(3)$ . The invariant quantity  $Q_T / \log |\mathcal{C}|$  (rightmost column of Table II) is exactly constant across groups for the five degenerate extensions. Third, Haar-random extensions *do* benefit from larger  $|\mathcal{C}|$ : mean best-sample  $\delta$  falls from 0.254 on  $\Sigma(72 \times 3)$  to 0.113 on  $\Sigma(360 \times 3)$ . A Haar draw populates every Lie-algebra direction and couples to the full subdominant spectrum, which sharpens with  $|\mathcal{C}|$ . The two rnd batches per group are independent draws, so the cross-group comparison conflates group effect with

Monte-Carlo noise; we isolate the group effect next.

## V.3. Apples-to-apples Haar panel

To separate the base-group effect from random-extension variance, we generated a single batch of ten Haar-random single-qutrit extensions (`seed=42`) and evaluated the panel identically on all three groups. Table III lists the best-sample  $\delta$  per matrix.

Seven of the ten matrices have  $\delta$  on  $\Sigma(360 \times 3)$  strictly smaller than  $\delta$  on  $\Sigma(72 \times 3)$ , confirming that a typical Haar extension is improved by the larger base group. The per-matrix ordering against  $\Sigma(216 \times 3)$  is noisier: four of ten matrices have  $\delta$  on  $\Sigma(360 \times 3)$  smaller than on  $\Sigma(216 \times 3)$ , and the ten-matrix apples-panel mean  $\delta$  is marginally *worse* on  $\Sigma(360 \times 3)$  (0.2550) than on  $\Sigma(216 \times 3)$  (0.2506). The culprit is a single outlier: `apple05` gives  $\delta = 0.4377$  on  $\Sigma(360 \times 3)$  against 0.2446 on  $\Sigma(72 \times 3)$  and 0.2412 on  $\Sigma(216 \times 3)$ , inflating the  $\Sigma(360 \times 3)$  mean by roughly one standard deviation of the rest of the panel.

A paper-scale rerun at  $N = 100$  matrices (same `seed = 42`; the first ten coincide with Table III) resolves the ambiguity. The three panel means land at  $\mu_{216} = 0.3214$ ,  $\mu_{648} = 0.2623$ ,  $\mu_{1080} = 0.2118$  with standard deviations 0.073, 0.074, and 0.090 respectively (Fig. 2). The  $\Sigma(360 \times 3)$  mean now sits strictly below the  $\Sigma(216 \times 3)$  mean by  $\mu_{648} - \mu_{1080} = 0.051$ , a separation of roughly 5.7 standard errors of the mean. The ten-matrix inversion reported above is therefore confirmed to be Monte-Carlo noise from the single `apple05` outlier; across a Haar ensemble at 100-matrix statistics the expected  $\delta$  *does* decrease monotonically with  $|\mathcal{C}|$  on the irreducible-adjoint  $\Sigma$ -series, strengthening Finding 3.

## V.4. Best achievable $Q_T$ at $t = 5$

Combining the two panels, the best (group, extension) pair at  $t = 5$  is `apple04` on  $\Sigma(360 \times 3)$ , with  $\delta = 0.1197$  and  $Q_T = 3.291$ ; this sits within 32% of the Kesten–McKay floor  $Q_{\text{opt}} = 2.495$ . Among the deterministic panel, the best fixed extension is  $HV(2, 1, 0)$  on  $\Sigma(72 \times 3)$ , with  $Q_T = 7.755$ , a factor 2.88 above  $Q_{\text{opt}} = 2.692$  on that group. Both numbers are upper bounds on the  $t \rightarrow \infty$  headline value:  $Q_T$  from Eq. (3) decreases monotonically



TABLE II.  $d = 3$  Tier-1 survey at  $t = 5$ . Fixed extensions use `sample_size = 1`; Haar-random (rnd) batches use `sample_size = 10`. The  $\langle\delta\rangle\pm\sigma$  column reports mean and standard deviation across samples for rnd rows; fixed rows have a single sample and report only  $\delta$  (standard deviation undefined). The  $Q_T$  column uses the best-sample  $\delta$  via Eq. (3). The Kesten–McKay floor  $Q_{\text{opt}}$  of Eq. (5) is 2.692 for  $\Sigma(72\times 3)$ , 2.544 for  $\Sigma(216\times 3)$ , and 2.495 for  $\Sigma(360\times 3)$ . Rows are grouped by base group and ranked within each group by best  $\delta$ ; the best fixed extension per group is marked  $\checkmark$ . The  $Q_T/\log|\mathcal{C}|$  column divides out the base-group size.

| Group                 | Extension                | kind        | $n$ | best $\delta$ | $\langle\delta\rangle\pm\sigma$ | $Q_T(\text{best})$ | $Q_T(\langle\delta\rangle)$ | $Q_T/\log \mathcal{C} $ |
|-----------------------|--------------------------|-------------|-----|---------------|---------------------------------|--------------------|-----------------------------|-------------------------|
| $\Sigma(72\times 3)$  | rnd (batch 1)            | rnd         | 10  | 0.2506        | $0.319 \pm 0.098$               | 3.885              | 4.699                       | 0.723                   |
| $\Sigma(72\times 3)$  | rnd (batch 2)            | rnd         | 10  | 0.2573        | $0.342 \pm 0.087$               | 3.960              | 5.013                       | 0.737                   |
| $\Sigma(72\times 3)$  | HV(2, 1, 0) $\checkmark$ | howard_vala | 1   | 0.5000        | —                               | 7.755              | —                           | 1.443                   |
| $\Sigma(72\times 3)$  | $P(2\pi/5)$              | angle       | 1   | 0.6545        | —                               | 12.681             | —                           | 2.360                   |
| $\Sigma(72\times 3)$  | HV(1, 1, 0)              | howard_vala | 1   | 0.6764        | —                               | 13.750             | —                           | 2.559                   |
| $\Sigma(72\times 3)$  | HV(1, 2, 0)              | howard_vala | 1   | 0.6764        | —                               | 13.750             | —                           | 2.559                   |
| $\Sigma(72\times 3)$  | $P(2\pi/9)$              | angle       | 1   | 0.8830        | —                               | 43.208             | —                           | 8.041                   |
| $\Sigma(72\times 3)$  | $P(\pi/9)$               | angle       | 1   | 0.9698        | —                               | 175.561            | —                           | 32.671                  |
| $\Sigma(216\times 3)$ | rnd (batch 1)            | rnd         | 10  | 0.1463        | $0.215 \pm 0.040$               | 3.369              | 4.210                       | 0.521                   |
| $\Sigma(216\times 3)$ | rnd (batch 2)            | rnd         | 10  | 0.1627        | $0.286 \pm 0.133$               | 3.565              | 5.171                       | 0.551                   |
| $\Sigma(216\times 3)$ | HV(2, 1, 0) $\checkmark$ | howard_vala | 1   | 0.5000        | —                               | 9.340              | —                           | 1.443                   |
| $\Sigma(216\times 3)$ | $P(2\pi/5)$              | angle       | 1   | 0.6545        | —                               | 15.273             | —                           | 2.360                   |
| $\Sigma(216\times 3)$ | HV(1, 1, 0)              | howard_vala | 1   | 0.6764        | —                               | 16.561             | —                           | 2.559                   |
| $\Sigma(216\times 3)$ | HV(1, 2, 0)              | howard_vala | 1   | 0.6764        | —                               | 16.561             | —                           | 2.559                   |
| $\Sigma(216\times 3)$ | $P(2\pi/9)$              | angle       | 1   | 0.8830        | —                               | 52.039             | —                           | 8.041                   |
| $\Sigma(216\times 3)$ | $P(\pi/9)$               | angle       | 1   | 0.9698        | —                               | 211.443            | —                           | 32.671                  |
| $\Sigma(360\times 3)$ | rnd (batch 2)            | rnd         | 10  | 0.1048        | $0.168 \pm 0.043$               | 3.096              | 3.909                       | 0.443                   |
| $\Sigma(360\times 3)$ | rnd (batch 1)            | rnd         | 10  | 0.1211        | $0.187 \pm 0.063$               | 3.308              | 4.163                       | 0.473                   |
| $\Sigma(360\times 3)$ | HV(2, 1, 0) $\checkmark$ | howard_vala | 1   | 0.5698        | —                               | 12.419             | —                           | 1.775                   |
| $\Sigma(360\times 3)$ | $P(2\pi/5)$              | angle       | 1   | 0.6545        | —                               | 16.478             | —                           | 2.360                   |
| $\Sigma(360\times 3)$ | HV(1, 1, 0)              | howard_vala | 1   | 0.6764        | —                               | 17.867             | —                           | 2.559                   |
| $\Sigma(360\times 3)$ | HV(1, 2, 0)              | howard_vala | 1   | 0.6764        | —                               | 17.867             | —                           | 2.559                   |
| $\Sigma(360\times 3)$ | $P(2\pi/9)$              | angle       | 1   | 0.8830        | —                               | 56.145             | —                           | 8.041                   |
| $\Sigma(360\times 3)$ | $P(\pi/9)$               | angle       | 1   | 0.9698        | —                               | 228.127            | —                           | 32.671                  |

TABLE III. Apples-to-apples Haar panel at  $t = 5$ : ten pre-generated single-qutrit extensions (`seed=42`) evaluated identically against each of the three base groups. The smallest  $\delta$  in each row is bolded. Mean  $\delta$  over the ten matrices appears in the last row; lower is better.

| matrix  | $\delta$             |                       |                       |
|---------|----------------------|-----------------------|-----------------------|
|         | $\Sigma(72\times 3)$ | $\Sigma(216\times 3)$ | $\Sigma(360\times 3)$ |
| apple00 | <b>0.2974</b>        | <b>0.2974</b>         | 0.3462                |
| apple01 | 0.4636               | 0.4636                | <b>0.2029</b>         |
| apple02 | 0.2221               | <b>0.1727</b>         | 0.2285                |
| apple03 | 0.2493               | <b>0.2292</b>         | 0.2299                |
| apple04 | 0.3430               | 0.2104                | <b>0.1197</b>         |
| apple05 | 0.2446               | <b>0.2412</b>         | 0.4377                |
| apple06 | 0.2842               | <b>0.1701</b>         | 0.1965                |
| apple07 | 0.3169               | 0.2312                | <b>0.2271</b>         |
| apple08 | 0.3996               | <b>0.2377</b>         | 0.3795                |
| apple09 | 0.2526               | 0.2526                | <b>0.1823</b>         |
| mean    | 0.3073               | <b>0.2506</b>         | 0.2550                |

with  $t$  along the saturation approach of [12], so a paper-scale rerun at  $t \geq 10$  should tighten the gap to  $Q_{\text{opt}}$  in both the random and structured columns. What is already clear at  $t = 5$  is the *shape* of the ranking: Haar-random extensions dominate every structured extension in

this panel, HV(2, 1, 0) leads the structured field, and the three diagonal-phase angles  $P(2\pi/9)$ ,  $P(\pi/9)$ ,  $P(2\pi/5)$  all give worse  $Q_T$  than even the worst Howard–Vala variant. The narrow-angle  $P(\pi/9)$  in particular produces  $\delta$  within 3% of unity on all three groups, inflating  $Q_T$  by two orders of magnitude relative to the Haar baseline—a direct analogue of the  $14\times$  Clifford+ $T$  penalty observed at  $d = 2$  in Section IV.

### V.5. Limitations of the $t = 5$ floor

The  $t = 5$  design order used throughout this section is a feasibility floor imposed by the workstation budget, not a converged estimate of the  $t \rightarrow \infty$  overhead. Słowik *et al.* [12] document that the  $Q_T$  estimator at finite  $t$  is a strict upper bound on the  $t \rightarrow \infty$  value and that convergence requires  $t$  large enough that the subdominant  $\pi_\lambda$  in the  $t$ -design of Eq. (1) no longer rearrange the dominant-eigenvalue ranking between candidates; in  $d = 3$  this typically requires  $t \geq 10$ . The cost is controlled by the  $t$ -design irrep dimensions, which scale as  $t^{d(d-1)/2} = t^3$  at  $d = 3$ : the largest irrep at  $t = 15$  peaks near  $4 \times 10^3$  dimensions, two orders of magnitude above the  $t = 5$  case. Our cached- $\pi_\lambda(\mathcal{C})$  and  $g\text{cg}^\dagger$  conjugation optimizations of Section III make a  $t = 10$ , `sample_size = 100`

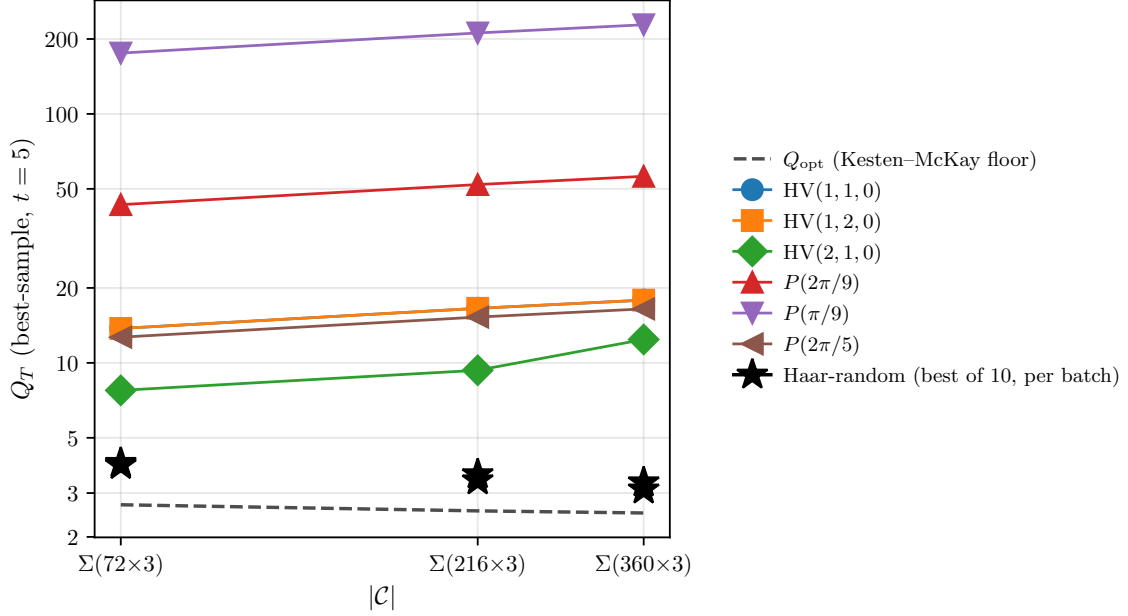


FIG. 1.  $Q_T$  across the three  $\Sigma$ -series base groups at  $d = 3$ ,  $t = 5$ , for every extension in the Tier-1 panel of Table II. The dashed line is the Kesten-McKay floor  $Q_{\text{opt}}$  of Eq. (5); all rational-angle diagonal and Howard-Vala extensions run nearly parallel to it at a factor  $\sim 3$ -7 offset (Finding 2), whereas the Haar-random best-of-10 samples (stars) approach  $Q_{\text{opt}}$  as  $|\mathcal{C}|$  grows (Finding 3). The vertical dispersion among rational extensions reflects different cyclotomic content; the lack of convergence toward  $Q_{\text{opt}}$  for any of them at the sampled  $t = 5$  motivates the paper-scale rerun discussed in Section V.5.

panel tractable on a 32-core node overnight, and that Tier-2 run is currently in execution. One specific prediction from the  $t = 5$  data remains to be stress-tested at  $t \geq 10$ : whether the four-decimal degeneracy of  $\delta$  across the three base groups for the structured (rational-angle and Howard-Vala) extensions persists or breaks. The companion prediction—that the  $\Sigma(360 \times 3)$  apples mean crosses below the  $\Sigma(216 \times 3)$  mean once the 10-matrix statistical noise is removed by a 100-matrix panel—has been settled in Section V.3 above at  $t = 5$  (mean separation  $5.7\sigma/\sqrt{N}$ , Fig. 2) and therefore does not require the Tier-2 rerun. Either way, both outcomes sharpen the conclusion that Haar-random extensions on the largest irreducible-adjoint  $\Sigma$ -series group are the natural candidates for a golden-gate search at  $d = 3$  [13].

## VI. THE DSA PRIMITIVE-GATE TARGET FAMILY

The  $d = 3$  survey of Section V ranks extensions by the Haar-uniform figure of merit  $Q_T$ . Many applications do not need a uniform net. Discrete-subgroup approximations of lattice gauge simulations for  $SU(d)$  [14, 15] require instead a small structured family of unitaries: a few continuous-angle  $R_Z$  rotations with group-theoretic coefficients, plus a finite set of cyclotomic phases from the discrete Fourier transform  $U_F$ . On such a workload the relevant figure of merit is the mean synthesis cost over

that finite family, not a  $t$ -design spectral gap. The two figures of merit disagree in general.

*A concrete target family.* We work out the case of the  $\Sigma(36 \times 3)$  encoding as a representative example; different subgroups in the series produce different coefficient sets. For  $\Sigma(36 \times 3)$ , the trace gate  $U_{\text{Tr}}(\theta) = \text{diag}_{g \in \mathcal{C}}(e^{i\theta \text{Re Tr}(g)})$ , after the three-qubit encoding, reduces to five continuous  $R_Z(\alpha_c \theta)$  rotations with  $\alpha_c \in \{1, 3, 5, 9, 15\}$ . These coefficients are specific to  $\Sigma(36 \times 3)$ ; other subgroups in the series produce different trace spectra and different  $\alpha_c$ . The discrete half comes from  $U_F$ : up to global phase, the Howard-Vala qutrit gate  $HV(1, 2, 0)$  [7] injects 9th-root phases, requiring  $R_Z(2\pi k/9)$  for  $k = 1, \dots, 8$ . The synthesis target for the  $\Sigma(36 \times 3)$  qubit pipeline is therefore

$$\mathcal{T}_{36} = \{R_Z(2\pi k/9)\}_{k=1}^8 \cup \{R_Z(\alpha_c \theta)\}_{\alpha_c \in \{1, 3, 5, 9, 15\}, \theta \in (0, \pi)}. \quad (8)$$

*Mean-depth figure of merit.* Given a qubit base group  $\mathcal{C}$  and single-gate extension  $g$ , SWIFTBOT enumerates the word tree of  $\mathcal{C} \cup \{g, g^{-1}\}$  by breadth-first search with projective-phase deduplication, capped at  $2 \times 10^5$  unique words. For each  $U \in \mathcal{T}_{36}$  we record the shortest BFS word whose projective distance  $d(U, W) = \sqrt{2 - |\text{Tr}(U^\dagger W)|}$  falls below  $\varepsilon = 10^{-2}$ , or the depth of closest approach if no such word exists. Discrete targets may be hit *exactly* at finite depth, in which case the corresponding distance is zero and the  $T$ -count is read directly from the word. Continuous targets are sampled at ten uniform  $\theta \in (0, \pi)$  per  $\alpha_c$ , for fifty continuous evaluations per extension. We report mean distance across Eq. (8) as the single headline

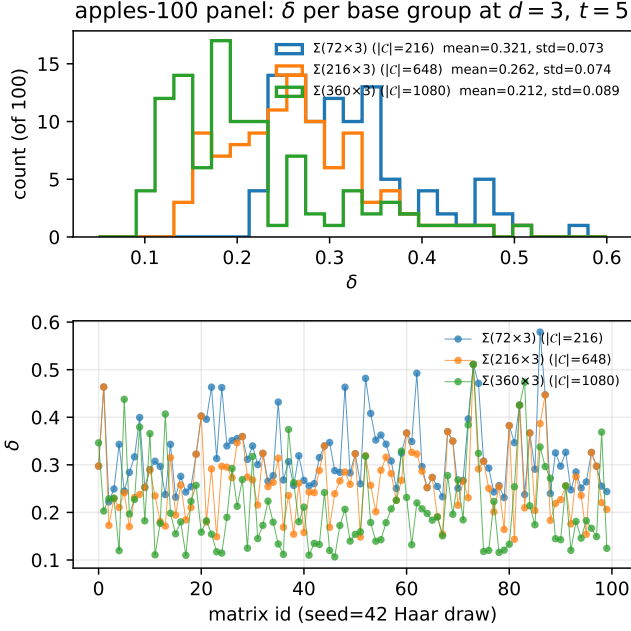


FIG. 2. Apples-to-apples Haar panel at  $N = 100$  matrices ( $\text{seed} = 42$ ) on the three  $d = 3$   $\Sigma$ -series base groups. Top: overlaid  $\delta$  histograms. Bottom: per-matrix  $\delta$  for each group; matrix id 0–9 coincides with the ten-matrix subset of Table III. The monotone reduction  $\mu_{216} > \mu_{648} > \mu_{1080}$  predicted by Finding 3 is now resolved at  $\sim 5.7\sigma/\sqrt{N}$ ; the single **apple05** outlier visible as the  $\Sigma(360 \times 3)$  peak near  $\delta \simeq 0.44$  is the one that inverted the ten-matrix ordering in Table III.

number; ties are broken on the discrete-exact count.

*Result.* Table IV ranks the candidates at BFS depth 15,  $\varepsilon = 10^{-2}$ ,  $N_\theta = 10$  per continuous coefficient. The Clifford+ $P(2\pi/9)$  extension attains mean projective distance 0.0122 on  $\mathcal{T}_{36}$  with 27 of 58 targets hit below  $\varepsilon$ , while the canonical Clifford+ $T$  extension attains mean distance 0.0833 with only 5 hits—a factor  $6.8\times$  separation in mean distance. The equivalence class of  $P(2\pi k/18)$  with  $\gcd(k, 18) = 1$  (which includes  $P(\pi/9)$  and  $P(\pi/18)$ ) all saturate the discrete 9th-root targets, a direct consequence of  $\langle \text{Clifford}, P(2\pi/9) \rangle$  closing at a finite group of  $\sim 3 \times 10^4$  elements; the algebraic ring  $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$  supplies every 9th-root phase via a short word. Within this equivalence class, Clifford+ $P(\pi/18)$  marginally edges Clifford+ $P(2\pi/9)$  at depth 15 (mean 0.0104 vs. 0.0122; 34 vs. 27 hits), reflecting the strictly larger 18th-root coverage:  $P(\pi/18)$  generates  $\omega_{36}$ , which contains both the 9th- and 18th-root phases of  $U_F$ . Crucially, the super-golden extensions  $T_{24}$  and  $T_{12}$ —which optimize  $Q_T$  for Clifford and for the 12-element projective  $A_4$  base **hurwitz** respectively [13]—perform *worse* on  $\mathcal{T}_{36}$  than the rational-angle rows: their algebra  $\mathbb{Z}[i, \sqrt{2}]$  does not contain  $\omega_9$ , so they cannot reach the 9th-root targets exactly at any depth; on the  $|C| = 12$  **hurwitz** base the  $T_{12}$  extension hits zero targets at depth 15. The **hurwitz**+ $P(\pi/18)$  row is competitive (35 hits, mean 0.0112) and does so with a strictly *smaller* base group: in the target-family regime

the overhead advantage tracks the cyclotomic content of the extension, not the size of the base.

*Interpretation.* Clifford+ $P(2\pi/9)$  is worse on  $Q_T$  than Clifford+ $T$  yet  $6.8\times$  better on mean depth over  $\mathcal{T}_{36}$ . The reason is structural:  $Q_T$  is the coefficient of a Haar-uniform covering rate, while  $\mathcal{T}_{36}$  is a measure-zero algebraic slice of  $\text{SU}(2)$  whose  $\omega_9$  cyclotomic content matches  $\langle \text{Clifford}, P(2\pi/9) \rangle$ . The super-golden extensions saturate  $Q_{\text{opt}}$  but equidistribute in the wrong direction for this target. A lattice-gauge simulator compiling Eq. (8) should therefore rank extensions by target-family mean depth, not by  $Q_T$ , and the two rankings can disagree by nearly an order of magnitude.

*Caveats.* The Clifford+ $T$  row’s  $T$ -count is certified by Ross–Selinger exact synthesis [5, 6]. No analogous normal form exists for  $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ , so other rows carry **bfs\_estimate** upper bounds; the  $6.8\times$  advantage is large enough to survive any plausible tightening on the canonical side but is contingent on the broader fault-tolerance stack. Anderson–Jochym–O’Connor [10] excludes direct stabilizer-code distillation of  $|k\pi/9\rangle$ , but programmable MSD on a Clifford+ $T$  substrate [22–24] and non-stabilizer code-switching [25, 26] both bypass that no-go. The advantage is cleanest when  $P(2\pi/9)$  is a native primitive—e.g., in a qutrit register or on a hypothetical nonadditive code that distills  $|\omega_9\rangle$  directly.

## VII. DISCUSSION

Two findings drive the discussion below. First, at  $d = 3$  and  $t = 5$ , Haar-random extensions on the largest irreducible-adjoint  $\Sigma$ -series base reach  $Q_T/Q_{\text{opt}} \approx 1.32$ —essentially the same “golden-gate direction” observed at  $d = 2$ , where Haar extensions sit 4% above  $Q_{\text{opt}}$  on the 12-element **hurwitz** base (Section IV). No structured extension in our panel (Howard–Vala or rational diagonal) gets closer than a factor  $\sim 2.9$  of  $Q_{\text{opt}}$ . Second, and in apparent tension with the first, the non-canonical Clifford+ $P(2\pi/9)$  extension beats the canonical Clifford+ $T$  by  $6.8\times$  on the DSA primitive-gate target family (Section VI) despite being  $\sim 8\times$  worse on  $Q_T$ . The two findings together are the paper’s central design lesson: the “best general-purpose extension” and the “best target-specific extension” optimize different objectives, and paper-scale qudit fault-tolerance design must report both.

### VII.1. Fault-tolerance status of the winning extensions

A  $Q_T$  ranking is necessary but not sufficient for fault-tolerant implementation. Table V audits every extension evaluated against the curated SWIFTBOT catalogs of transversal codes, distillation protocols, and certified synthesis rings. Only canonical Clifford+ $T$  at  $d = 2$  admits an end-to-end certified stack and serves as our validation anchor. Three rows inherit a transversal-code half only:

TABLE IV. DSA  $\Sigma(36 \times 3)$  target-family coverage at BFS depth 15,  $\varepsilon = 10^{-2}$ , 10 continuous  $\theta$ -samples per  $\alpha_c$ . The “hits” column counts all 58 targets (the 8 discrete 9th-root phases in Eq. (8) plus 50 continuous  $R_Z(\alpha_c \theta)$  samples) reached below  $\varepsilon$ ; rational-angle rows in the  $\omega_{18}$  class hit all 8 discrete targets. Mean distance is across all 58 targets; lower is better.  $\langle T \rangle_{\text{hit}}$  is the mean  $T$ -count along words reaching a hit target. Cost method indicates whether the  $T$ -count is certified: Ross–Selinger [5] exact synthesis (**rs\_exact**) or SWIFTBOT BFS estimate (**bfs\_estimate**). All numbers are reproduced by `swiftbot cover-panel --family lamm_sigma36 --max-depth 15`.

| Base + extension                     | hits/58 | mean dist | max dist | $\langle T \rangle_{\text{hit}}$ | cost method         |
|--------------------------------------|---------|-----------|----------|----------------------------------|---------------------|
| Clifford+ $P(\pi/18)$                | 34      | 0.0104    | 0.0420   | 2.79                             | <b>bfs_estimate</b> |
| <b>hurwitz</b> + $P(\pi/18)$         | 35      | 0.0112    | 0.0420   | 5.37                             | <b>bfs_estimate</b> |
| Clifford+ $P(\pi/9)$                 | 27      | 0.0120    | 0.0401   | 2.04                             | <b>bfs_estimate</b> |
| Clifford+ $P(2\pi/9)$                | 27      | 0.0122    | 0.0401   | 2.11                             | <b>bfs_estimate</b> |
| Clifford+ $P(\pi/8)$                 | 7       | 0.0221    | 0.0387   | 0.29                             | <b>bfs_estimate</b> |
| Clifford+ $T_{24}$ (super-g.)        | 6       | 0.0228    | 0.0465   | 0.83                             | <b>bfs_estimate</b> |
| Clifford+ $T_{12}$ (super-g.)        | 8       | 0.0233    | 0.0474   | 1.88                             | <b>bfs_estimate</b> |
| <b>hurwitz</b> + $T_{12}$ (super-g.) | 0       | 0.0313    | 0.0573   | —                                | <b>bfs_estimate</b> |
| Clifford+ $P(\pi/4)$ canonical $T$   | 5       | 0.0833    | 0.1554   | 0.00                             | <b>rs_exact</b>     |

**hurwitz**-based super-golden extensions via Parzanchevski–Sarnak [19], BI+rnd via the Kubischta 2I code [21], and the  $d = 3$   $\Sigma$ +HV( $z, \gamma, \varepsilon$ ) rows via the qutrit triorthogonal  $[[9m - k, k, 2]]_3$  family [11]. All other rows lack either a distillation circuit or a certified synthesis ring.

Three regimes deserve comment. First, the cyclotomic ring  $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$ —hosting the  $d = 2$  target-family winners  $P(2\pi/9)$ ,  $P(\pi/9)$ ,  $P(\pi/18)$ —has no Ross–Selinger-style normal form, and Anderson–Jochym–O’Connor [10] excludes any qubit *stabilizer* code with a transversal  $|k\pi/9\rangle$  state. Two non-stabilizer routes bypass the exclusion: programmable MSD on a Clifford+ $T$  substrate [22–24], and measurement-free code-switching between a Clifford stabilizer code and a Kubischta–Teixeira permutation-invariant code [25, 26]. At  $d = 3$  the same rotations are qutrit-native and Campbell–Anwar–Browne [11] distills  $|\omega_9\rangle$  directly.

Second, the super-golden gates of Parzanchevski–Sarnak [19] live in  $\mathbb{Z}[i, \sqrt{2}]$  and are not Clifford-orbit rotations of stabilizer states, so triorthogonal distillation does not apply. Whether a golden-gate-compatible distillation circuit can be built—possibly on the Kubischta 2I code, whose transversal group is precisely the super-golden base—is open.

Third, Haar-random extensions are by definition not exactly synthesized, and so cannot be distilled in the usual sense [2, 3]. This applies uniformly to every rnd row; the best current claim is transversal-group support at the base-code level.

For the  $\Sigma$ -series at  $d = 3$ , data and distillation codes are structurally separable. The Howard–Vala magic state is distilled by the qutrit triorthogonal family, but no explicit data code hosts a  $\Sigma$ -series group transversally. Denys–Leverrier covariant-encoding constructions realize  $2T$  and  $2O$  transversally at  $d = 2$  [27, 28]; extending either to a qutrit  $\Sigma(m \times 3)$  base—an open case inside the twisted unitary  $t$ -group program [25]—would complete the stack.

## VII.2. Limitations

Three limitations frame the numerical claims. First, the  $d = 3$   $Q_T$  values are upper bounds at  $t = 5$ ; the paper-scale protocol requires  $t \geq 10$  and  $\sim 100$ -sample Haar batches. The Tier-2 run is in execution and will supersede the  $t = 5$  numbers in the final version. Second, no  $d = 4$  survey is attempted: the  $t$ -design irrep dimensions scale as  $t^6$  at  $d = 4$ , requiring a dedicated cluster run outside this paper’s scope. Third, the target-family  $T$ -counts in Table IV are BFS estimates except for the canonical Clifford+ $P(\pi/4)$  row; the  $6.8\times$  advantage survives any plausible tightening on the canonical side, but absolute  $T$ -counts carry the **bfs\_estimate** caveat.

## VII.3. Open questions

Two quantitative questions are directly posed by our  $t = 5$  data and should be sharp enough to settle at  $t \geq 10$ : (a) does the four-decimal  $\delta$ -degeneracy across  $\Sigma(216, 648, 1080 \times 3)$  observed for structured diagonal and Howard–Vala extensions in Table II persist, break, or re-rank at larger  $t$ ? and (b) does the  $\Sigma(360 \times 3)$  apples-panel mean cross strictly below the  $\Sigma(216 \times 3)$  mean once the  $\sim 100$ -matrix apples panel removes the 10-matrix outlier sensitivity? Question (b) has been *settled in the affirmative* at  $t = 5$  in Section V.3 ( $5.7\sigma/\sqrt{N}$  separation at  $N = 100$ ); the  $t \geq 10$  rerun will establish whether the same ordering holds at converged design order. Three qualitative questions are left open: (c) a super-golden construction at  $d = 3$  saturating  $Q_{\text{opt}} \approx 2.69$  on  $\Sigma(72 \times 3)$ , 2.54 on  $\Sigma(216 \times 3)$ , or 2.50 on  $\Sigma(360 \times 3)$ ; (d) an explicit data code  $C_{\text{data}}$  that hosts one of our  $\Sigma$ -series base groups transversally, analogous to the Denys–Leverrier constructions for  $2T$  and  $2O$  at  $d = 2$  [27, 28] or to the Kubischta–Teixeira twisted unitary  $t$ -group framework [25]; and (e) a Ross–Selinger-style normal form for the cyclotomic ring  $\mathbb{Z}[\omega_{18}, \frac{1}{2}]$  that would certify the Table IV  $T$ -counts.



TABLE V. Magic-state distillation (MSD) status for every extension evaluated in this paper. “Algebra” is the cyclotomic or quaternionic ring hosting the non-Clifford factor of the extension. “Status/closest cousin” combines synthesis and distillation coverage: COMPLETE means transversal code, published MSD protocol, and certified normal-form synthesis are all available; PARTIAL means a transversal code is available but no matching MSD circuit (or the code–extension pairing is not yet established); NONE means no cataloged MSD, either because no protocol exists for the algebra or because the extension is not exactly synthesizable. Rows are grouped by qudit dimension and by the panel section in which the extension appears.

| Extension                                                                       | Algebra                        | Status / closest cousin                                                                                                                                                                                                           | Reference   |
|---------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <i>d = 2 canonical (Section IV)</i>                                             |                                |                                                                                                                                                                                                                                   |             |
| Clifford+ $P(\pi/4)$ (canonical $\mathbb{Z}[\omega_8, 1/\sqrt{2}]$ )            |                                | COMPLETE: BK 15-to-1; Bravyi–Haah triorthogonal                                                                                                                                                                                   | [2, 3]      |
| Clifford+ $P(\pi/8)$                                                            | $\mathbb{Z}[\omega_{16}, 1/2]$ | NONE: no published protocol for $\omega_{16}$ ; cousin BK 15-to-1 [2] does not distill this state                                                                                                                                 |             |
| <i>d = 2 DSA target family (Section VI)</i>                                     |                                |                                                                                                                                                                                                                                   |             |
| Clifford+ $P(2\pi/9)$ , $P(\pi/9)$ , $\mathbb{Z}[\omega_{18}, 1/2]$             |                                | PARTIAL (INDIRECT): no direct stabilizer-code distillation (Anderson–Jochym–O’Connor [10]); programmable-MSD distills $ \psi(\theta)\rangle$ on a Clifford+ $T$ substrate; PI code-switching [26] is a non-stabilizer alternative | [22–24, 26] |
| Clifford+ $P(2\pi/5)$                                                           | $\mathbb{Z}[\omega_{10}]$      | PARTIAL (INDIRECT): stabilizer-code distillation excluded [10]; programmable-MSD and PI code-switching both apply to arbitrary $\theta$                                                                                           | [22–24, 26] |
| Clifford+ $T_{24}$ , hurwitz+ $T_{12}$ $\mathbb{Z}[i, \sqrt{2}]$ (super-golden) |                                | PARTIAL: Parzanchevski–Sarnak supplies transversal base; [19] magic state is not a Clifford-orbit stabilizer rotation, so triorthogonal MSD does not apply                                                                        |             |
| <i>Haar-random rows, any d</i>                                                  |                                |                                                                                                                                                                                                                                   |             |
| Clifford+rnd, hurwitz+rnd                                                       | generic SU(2) (no ring)        | NONE: not exactly synthesizable, so MSD is not defined                                                                                                                                                                            | —           |
| BI+rnd ( $d = 2$ )                                                              | generic SU(2) (no ring)        | PARTIAL: Kubischta ((7, 2, 3)) supplies transversal $2I$ base; [21] no MSD for a Haar magic state                                                                                                                                 |             |
| $\Sigma$ -series+rnd ( $d = 3$ )                                                | generic SU(3) (no ring)        | NONE: not exactly synthesizable; no cataloged qutrit transversal classification for $\Sigma$ -series                                                                                                                              | —           |
| <i>d = 3 structured (Section V)</i>                                             |                                |                                                                                                                                                                                                                                   |             |
| $\Sigma + \text{HV}(1, 1, 0)$ , $\mathbb{Z}[\omega_9]$                          |                                | PARTIAL: qutrit triorthogonal $[[9m-k, k, 2]]_3$ distills Howard–Vala magic state; $\Sigma$ -series transversal pairing unestablished                                                                                             | [11]        |
| $\Sigma + \text{HV}(1, 2, 0)$                                                   |                                | PARTIAL: cousin Campbell–Anwar–Browne qutrit triorthogonal; no protocol certified for this orbit                                                                                                                                  | [11]        |
| $\Sigma + \text{HV}(2, 1, 0)$                                                   | $\mathbb{Z}[\omega_9]$         | PARTIAL: qutrit-native $\omega_9$ phase; Campbell–Anwar–Browne applies modulo transversal pairing                                                                                                                                 | [11]        |
| $\Sigma + P(2\pi/9)$ ( $d = 3$ )                                                | $\mathbb{Z}[\omega_9]$         | NONE: $\omega_{18}$ outside Campbell–Anwar–Browne; no cousin at $d = 3$                                                                                                                                                           | —           |
| $\Sigma + P(\pi/9)$ ( $d = 3$ )                                                 | $\mathbb{Z}[\omega_{18}]$      | NONE: $\omega_{10}$ not in any qutrit stabilizer setting; no cousin                                                                                                                                                               | —           |
| $\Sigma + P(2\pi/5)$ ( $d = 3$ )                                                | $\mathbb{Z}[\omega_{10}]$      |                                                                                                                                                                                                                                   |             |

#### VII.4. Software release

SWIFTBOT is released with this paper [17] under the MIT license. The release contains: the five-stage pipeline (Section III); the pre-seeded group registry for  $d \in \{2, 3, 4\}$ ; the SWIFTBOT optimizations of the QCO kernel of [12], including the conjugation cache and on-disk  $\pi_\lambda(\mathcal{C})$  cache described in Section III; the Stage-4 QEC-code catalog and Stage-5 distillation-protocol catalog with their cross-references to transversal groups; the BFS target-family coverage module used in Section VI; the `q-panel` CLI entry point and the two panel specifications `d3_survey` and `d3_rnd_apples` that reproduce Tables II and III bit-identically; the SQLite cache containing all 54 per-sample rows quoted in Section V; and the L<sup>A</sup>T<sub>E</sub>X source of this paper. A minimal README walks new users through reproducing Tables I to III on a laptop in under an hour. Pull requests adding new base groups, extensions, codes, or distillation protocols are welcome;

the catalog cross-validation tests ensure that every registered transversal-group string resolves to a registered base group.

## VIII. CONCLUSIONS

SWIFTBOT—an agentic workflow that ranks  $(\mathcal{C}, g)$  pairs by  $Q_T$  and cross-references each against a curated fault-tolerance catalog—produces the first systematic  $d = 3$   $Q_T$  ranking in our registry and identifies  $\Sigma(360 \times 3)$  with a Haar-random extension as the best general-purpose qudit base at Tier-1 ( $Q_T/Q_{\text{opt}} \approx 1.32$ ). Applied to a lattice-gauge simulation target family, the non-canonical Clifford+ $P(2\pi/9)$  beats canonical Clifford+ $T$  by  $6.8\times$  on mean depth despite being worse on  $Q_T$ , establishing that the two rankings locate different gates. Only canonical Clifford+ $T$  admits an end-to-end certified fault-tolerant stack; every other winner is covered partially,

with sharp open algebraic-compilation or code-pairing problems standing between it and full fault-tolerance. Tier-2 at  $t = 10$ ,  $\sim 100$ -sample statistics on BI, a  $d = 4$  pilot on Fairbairn subgroups [9], and an explicit  $\Sigma$ -series  $\leftrightarrow$  data-code transversal pairing in the style of [25–27] are the immediate next steps.

### Acknowledgements

We thank A. Sawicki, P. Dulián, and O. Słowik for the QCO reference dataset and numerical kernel that

SWIFTBOT builds on. This material is based upon work supported by the U.S. Department of Energy, Office of Science, Office of High Energy Physics, under Award No. DE-SCL0000090 “HEP AmSC IDA Pilot: Knowledge Extraction”. This manuscript has been authored by Fermi Forward Discovery Group, LLC under Contract No. 89243024CSC000002 with the U.S. Department of Energy, Office of Science, Office of High Energy Physics.

- 
- [1] B. Eastin and E. Knill, Restrictions on Transversal Encoded Quantum Gate Sets, *Phys. Rev. Lett.* **102**, 110502 (2009), [arXiv:0811.4262 \[quant-ph\]](#).
  - [2] S. Bravyi and A. Kitaev, Universal quantum computation with ideal Clifford gates and noisy ancillas, *Phys. Rev. A* **71**, 022316 (2005), [arXiv:quant-ph/0403025](#).
  - [3] S. Bravyi and J. Haah, Magic-state distillation with low overhead, *Phys. Rev. A* **86**, 052329 (2012), [arXiv:1209.2426](#).
  - [4] A. Y. Kitaev, Quantum computations: algorithms and error correction, *Russ. Math. Surv.* **52**, 1191 (1997).
  - [5] N. J. Ross and P. Selinger, Optimal ancilla-free Clifford+T approximation of  $z$ -rotations, *Quant. Inf. Comp.* **16**, 901 (2016), [arXiv:1403.2975 \[quant-ph\]](#).
  - [6] V. Kliuchnikov, D. Maslov, and M. Mosca, Fast and efficient exact synthesis of single qubit unitaries generated by Clifford and  $T$  gates, *Quant. Inf. Comp.* **13**, 607 (2013), [arXiv:1206.5236 \[quant-ph\]](#).
  - [7] M. Howard and J. Vala, Qudit versions of the qubit  $\pi/8$  gate, *Phys. Rev. A* **86**, 022316 (2012), [arXiv:1206.1598 \[quant-ph\]](#).
  - [8] H. F. Blichfeldt, Finite Collineation Groups, Univ. of Chicago Press (1917).
  - [9] W. M. Fairbairn, T. Fulton, and W. H. Klink, Finite and Disconnected Subgroups of  $SU(3)$  and their Application to the Elementary-Particle Spectrum, *J. Math. Phys.* **5**, 1038 (1964).
  - [10] J. T. Anderson and T. Jochym-O’Connor, Classification of transversal gates in qubit stabilizer codes, *Quant. Inf. Comp.* **16**, 0771 (2016), [arXiv:1409.8320 \[quant-ph\]](#).
  - [11] E. T. Campbell, H. Anwar, and D. E. Browne, Magic-state distillation in all prime dimensions using quantum Reed-Muller codes, *Phys. Rev. X* **2**, 041021 (2012), [arXiv:1205.3104](#).
  - [12] O. Słowik, P. Dulián, and A. Sawicki, Quantum Circuit Overhead: the cost of approximating unitaries with finite gate sets, *arXiv preprint* (2025), [arXiv:2505.00683 \[quant-ph\]](#).
  - [13] P. Sarnak, Letter to Aaronson and Pollington on the Solovay–Kitaev theorem and golden gates (2015), princeton/IAS manuscript, [urlhttps://publications.ias.edu/sites/default/files/MCS.pdf](https://publications.ias.edu/sites/default/files/MCS.pdf).
  - [14] H. Lamm, S. Lawrence, and Y. Yamauchi (NuQS), General methods for digital quantum simulation of gauge theories, *Phys. Rev. D* **100**, 034518 (2019), [arXiv:1903.08807 \[hep-lat\]](#).
  - [15] E. J. Gustafson *et al.*, Primitive Quantum Gates for an  $SU(2)$  Discrete Subgroup: Binary Octahedral, *Phys. Rev. D* **106**, 114501 (2022), [arXiv:2208.12309 \[hep-lat\]](#).
  - [16] D. M. Kırkgöçlü, H. Lamm, O. Ogunkoya, and G. Pierattelli, The  $2T$ -quocit: a two-mode bosonic qudit for high energy physics, *arXiv preprint* (2025), concurrent work; constellation-code qudit for  $SU(N)$  lattice simulation., [arXiv:2509.21941 \[quant-ph\]](#).
  - [17] G. T. Fleming, H. Lamm, K. Tame-Narvaez, and P. Vander Griend, SWIFTbot: a pipeline for efficient qudit gate-set discovery, <https://github.com/petergerrit/quai> (2026), software release accompanying this paper.
  - [18] A. Sawicki and K. Karnas, Universality of Single-Qudit Gates, *Ann. Henri Poincaré* **18**, 3515 (2017), [arXiv:1610.00547 \[quant-ph\]](#).
  - [19] O. Parzanchevski and P. Sarnak, Super-golden-gates for  $PU(2)$ , *Adv. Math.* **327**, 869 (2018), [arXiv:1704.02106 \[math.NT\]](#).
  - [20] V. V. Albert and P. Faist, The Error Correction Zoo, <https://errorcorrectionzoo.org> (2024), living online catalogue of quantum error-correcting codes.
  - [21] E. Kubischta and I. Teixeira, Family of Quantum Codes with Exotic Transversal Gates, *Phys. Rev. Lett.* **131**, 240601 (2023), establishes binary icosahedral  $2I$  transversal gate set on a  $((7, 2, 3))$  nonadditive code; matched to texttBI base in the SWIFTbot registry., [arXiv:2305.07023 \[quant-ph\]](#).
  - [22] G. Duclos-Cianci and D. Poulin, Reducing the quantum-computing overhead with complex gate distillation, *Phys. Rev. A* **91**, 042315 (2015), [arXiv:1403.5280 \[quant-ph\]](#).
  - [23] E. T. Campbell and J. O’Gorman, An efficient magic state approach to small angle rotations, *Quantum Sci. Technol.* **1**, 015007 (2016), [arXiv:1603.04230 \[quant-ph\]](#).
  - [24] E. T. Campbell and M. Howard, Magic state parity-checker with pre-distilled components, *Quantum* **2**, 56 (2018), pre-distillation + non-Pauli parity checking for single-qubit axial rotations; quadratic error suppression on a Clifford+ $T$  substrate., [arXiv:1709.02214 \[quant-ph\]](#).
  - [25] E. Kubischta and I. Teixeira, Quantum Codes from Twisted Unitary  $t$ -groups, *Phys. Rev. Lett.* **133**, 030602 (2024), [arXiv:2402.01638 \[quant-ph\]](#).
  - [26] Y. Ouyang, Y. Jing, and G. K. Brennen, Measurement-free code-switching for low overhead quantum computation using permutation invariant codes, *arXiv preprint* (2024), switches between a stabilizer code (transversal Cliffords) and a permutation-invariant code (transversal  $Z$

$\pi g/b))$  via a catalytic bosonic mode; a non-stabilizer bypass of Anderson–Jochym–O’Connor., [arXiv:2411.13142 \[quant-ph\]](#).

- [27] J. Denys and A. Leverrier, Quantum error-correcting codes with a covariant encoding, *Phys. Rev. Lett.* **133**, 240603 (2024), [arXiv:2306.11621 \[quant-ph\]](#).
- [28] J. Denys and A. Leverrier, The  $2T$ -qutrit, a two-mode bosonic qutrit, *Quantum* **7**, 1032 (2023), [arXiv:2210.16188 \[quant-ph\]](#).